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Least squares 2D bi-cubic spline approximation: Theory and applications

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ABSTRACT

Smooth surface approximation plays an important role in many applications. As an extension of the 2D bi-cubic spline interpolation, we propose the least squares 2D bi-cubic spline approximation (LS-BICSA). LS-BICSA, generally applicable to a set of irregular data, is formulated by the pure bi-cubic spline functions. The method is flexible because the user can impose the appropriate continuity and differentiability conditions at the boundary points using the available constrained least squares theory. Two scenarios having two and four knots at the boundary points are described in this paper. Two examples are presented to illustrate the theory. The first example approximates a known mathematical function on an irregular grid. The second example uses a real data set to approximate the geoid height. LS-BICSA can thus provide reliable results for many geoscience applications where it is required to approximate the function values using a smooth spline surface in the least squares sense.

1. Introduction

In many science and engineering applications, fitting a curve in one-dimensional (1D) space or a surface in two-dimensional (2D) space to a set of randomly scattered data points is a commonly encountered problem. To estimate the unknown function values of known data points, two methods are usually used: interpolation and approximation. Estimating the unknown coefficients of a polynomial passing through m points (known as knots) is referred to as interpolation. Approximation however requires that the polynomial gets sufficiently close to the knots [1]. The nature of the problem at hand usually determines to use either interpolation or approximation. A general statement may however hold true. In the case that the data are contaminated with random noise, approximation provides more accurate results than the pure interpolation method.

There is an alternative method to the classical polynomial interpolation, the spline interpolation. A spline function is a piecewise polynomial interpolation such that these polynomial segments are joined together at the knots under some continuity conditions [2–4]. The spline interpolation is preferred over the classical polynomial interpolation due to the following two reasons: 1) The interpolation error at intermediate points can be made sufficiently small even if one uses low order polynomials. 2) Spline interpolation can avoid the problem of Runge's phenomenon [5]. The Runge's phenomenon is the classic illustration of polynomial interpolation non-convergence [6]. It demonstrates the error or problem of high oscillation at the edges of function occurring in high order polynomial interpolations.

In general, spline polynomials can be made by an arbitrary degree. Examples include linear splines, quadratic splines and cubic splines. Splines of order p will usually have continuous derivatives up to order $p-1$ at the boundaries of the joined curves. In the third-order polynomials, called cubic splines, the slope (the first derivative) and the curvature (the second derivative) are continuous at the knots [7–9]. Cubic spline becomes a curve fitting problem for 1D data. We however need to solve a surface fitting problem for 2D data [10–12]. An important method in surface fitting in 2D is the so-called bi-cubic interpolation.

Bi-cubic interpolation is a form of cubic interpolation in two dimensions consisting of third-order polynomial pieces in grid squares. This interpolation problem consists of 16 unknown coefficients and four known knots, four known first derivatives in the u -direction, four known first derivatives in the v -direction and four known second derivatives of cross uv . Therefore we must have the function values and the first and second derivatives known at the knots. This method has zero redundancy and the interpolation surface can simply be obtained by a linear system of equations. This surface will then pass through each knot. Such pure and perfect interpolation can be used only if there are no errors in the function values. This turns out not to be the case for many real measurement set-ups. Approximation is thus preferred over interpolation.

An approximating polynomial does not necessarily pass through the knots. It is only required that the polynomial gets as close as possible to the knots. This can be achieved by the least squares method, which minimizes the sum of the squared errors [13–15]. This paper presents

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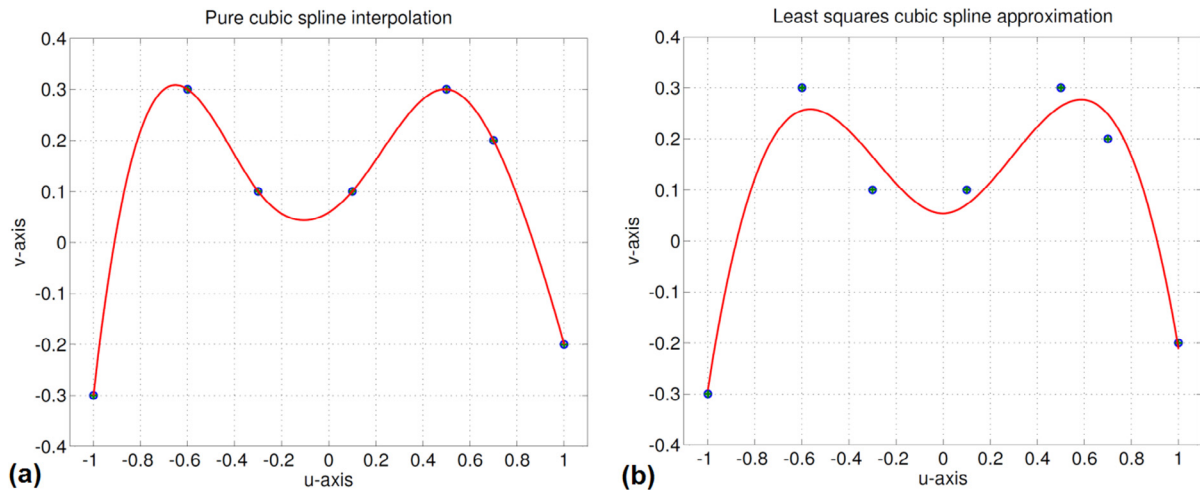


Fig. 1. Pure cubic spline interpolation, passing through knots (a), least squares cubic spline approximation, getting close to knots (b).

the least squares 2D bi-cubic spline approximation (LS-BICSA) theory to the surface fitting problem in the 2D space. This method often takes redundant measurements to enhance the precision and to check for the presence of measurement gross errors. Increasing the number of observations will also automatically lead to the use of small surface patches and hence it enhances the accuracy of the approximation method. Therefore, there are more equations than the minimum required. Bi-cubic spline interpolation is however used only for interpolating data points on a 2D regular grid. Although for the 1D spline interpolation method data on an irregular grid can be used, a set of grid points on a regular rectangular region is required for the 2D case. The LS-BICSA does not necessarily require a set of regular grid knots because the connecting knots are defined by the user. Also this method does not require the function values to be known at the boundary of the pieces. One just needs to impose the continuity conditions along and across a given direction at the boundary points of the surface patches. We thus may include a few hard constraints into the functional model. The solution can be sought by solving a constrained least squares problem [15].

In this paper we will follow to approximate a 2D surface by using the pure cubic splines. The approximation can likely be achieved by employing other basis functions such as B-splines [11,16–21], Bézier basis splines [22,23], radial basis functions [24], thin plate splines [25], T-splines [26,27], bivariate splines [28], triangular B-splines [29]. These basis functions usually impose the continuity constraints in an implicit form. A more elegant manner is that the constraints are defined by the user in an explicit form. Such explicitly imposed constraints make the proposed method flexible because the appropriate constraints can be identified and implemented by the user.

The interpolation and approximation problems addressed in the present contribution can also be tackled in a statistical manner using the method of best linear unbiased prediction (BLUP). This is basically because any interpolation, extrapolation and approximation problem using the basis functions such as polynomials can be treated as a linear unbiased prediction (LUP). A predictor is the outcome of a random vector having a mean that is a linear function of unknown parameters x . In particular, the stochastic properties of such a random vector should be specified. In many practical problems, this random vector is assumed to be stochastically correlated to the original observations y . By employing the minimum mean squared error (MMSE) criterion, such prediction becomes the best predictor. For further information on the theory and applications of the BLUP, we may refer to Refs. [30,31].

The remaining parts of this paper are organized as follows. Section 2 describes the difference between interpolation and approximation methods. We then provide an overview of the bi-cubic spline

interpolation method. The theoretical background of the least squares 2D bi-cubic spline approximation (LS-BICSA) is then provided in Section 3. The theory of constrained least squares is used to impose the continuity conditions to the functional model. To better explain the required number of hard constraints, two scenarios are proposed. In the first scenario, the surface patches are connected together at only two boundary points and the continuity conditions are imposed at these boundary points. In the second scenario, the patches are connected together at four boundary points. We then present numerical examples to show the efficacy of the proposed method in Section 4. The data set used is the geoid undulation obtained from the Global Geopotential Model EIGEN-6C4. Geoid height data of a high-resolution grid of $1' \times 1'$, covering an arbitrary area of $1^\circ \times 1^\circ$, is used in this study. The least squares residuals of the approximated surface are then calculated under 5 different patch sizes for the above two scenarios. Section 5 provides a summary and conclusion.

2. Spline interpolation and approximation

This section presents a few basic concepts on the use of cubic spline interpolation and approximation for the 1D and 2D datasets.

2.1. 1D cubic spline interpolation and approximation

Two methods of polynomial interpolation and spline interpolation are often used in interpolating problems. Polynomial interpolation is a global interpolation of which a single equation is fitted to all control points. This equation is usually a high order and smooth polynomial. The higher the order of the polynomial is, the smoother the function will be. A spline function is however a polynomial interpolation in a piecewise method. Each piece of data (between two adjacent knots in 1D data) is interpolated by the polynomials. These polynomial segments are low order polynomials and needed to join together in the knots under some continuity conditions [3,7,32]. Spline interpolation is widely used because it avoids the Runge's phenomenon [5,6].

The use of 1D polynomial fit using cubic splines is a curve fitting problem. For a given dataset, an interpolating polynomial must pass through each control point (Fig. 1a), whereas an approximating polynomial does not necessarily pass through its defining control points (Fig. 1b). In this figure the interpolating polynomial consists of six pieces of cubic polynomials connected to each other, whereas the approximating polynomial consists of two pieces, one defined on $[-1, 0]$ and the other on $[0, 1]$. For further information on the theory and application of least squares cubic spline approximation one may refer to [33].

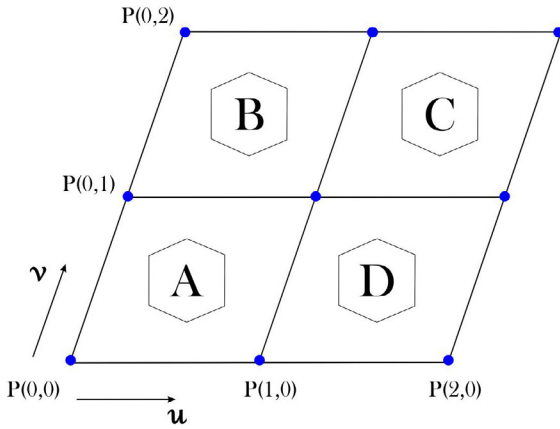


Fig. 2. Unit square patches with their known knots on a regular grid, applicable to 2D bi-cubic spline interpolation.

2.2. 2D Bi-cubic spline interpolation

A spline surface in 2D consists of a set of polynomial pieces (surface patches) that are smoothly connected together by some continuity conditions at their boundaries [2,3,7]. These pieces are sections defined by each grid squares in 2D data. The 2D spline polynomials can also be defined by any order, such as first-order polynomials, i.e., linear splines, which have only to be continuous at their knots. Second-order polynomials are quadratic splines, which have continuous slopes at the knots, but their curvatures are likely discontinuous. For the third-order polynomials, i.e., cubic splines, both the slopes and curvatures are continuous at the knots of the grid squares [7–9].

The theory of bi-cubic spline interpolation is now presented. Bi-cubic interpolation is a cubic interpolation in two dimensions. It consists of third-order polynomial pieces in grid squares [34,35]. Considering the data set of four unit squares patched together (Fig. 2), bi-cubic spline interpolation assigns a third-order 2D polynomial to each unit. It is of the form

$$f(u, v) = \sum_{i=0}^3 \sum_{j=0}^3 a_{ij} u^i v^j \quad (1)$$

which holds for the surface patch A in Fig. 2. Similar cubic polynomials can also be defined for patches B, C and D.

This interpolation problem consists of 16 unknown coefficients a_{ij} and four known knots (four equations) in each unit. For a pure interpolation problem, the number of known equations must be equal to the number of coefficients within each surface patch. We thus have to suppose that the first and second derivatives of the four corners are also known. Suppose the function values f , the first derivatives f_u, f_v and the cross derivatives f_{uv} are known at corner knots (0,0), (1,0), (0,1) and (1,1) of the unit square A in Fig. 2. Therefore, in total 16 boundary conditions (equations) are to be fulfilled to estimate the unknown coefficients of each polynomial. They are four function values at the corner knots as

$$\begin{aligned} f(0, 0) &= a_{00} \\ f(1, 0) &= a_{00} + a_{10} + a_{20} + a_{30} \\ f(0, 1) &= a_{00} + a_{01} + a_{02} + a_{03} \\ f(1, 1) &= \sum_{i=0}^3 \sum_{j=0}^3 a_{ij} \end{aligned} \quad (2)$$

In a similar manner, the eight tangent vectors, the first derivatives, at the corner knots (two at each point in the u and v directions) are of the form

$$\begin{aligned} f_u(0, 0) &= a_{10} \\ f_u(1, 0) &= a_{10} + 2a_{20} + 3a_{30} \\ f_u(0, 1) &= a_{10} + a_{11} + a_{12} + a_{13} \\ f_u(1, 1) &= \sum_{i=1}^3 \sum_{j=0}^3 i a_{ij} \\ \text{and} \\ f_v(0, 0) &= a_{01} \\ f_v(1, 0) &= a_{01} + a_{11} + a_{21} + a_{31} \\ f_v(0, 1) &= a_{01} + 2a_{02} + 3a_{03} \\ f_v(1, 1) &= \sum_{i=0}^3 \sum_{j=1}^3 j a_{ij} \end{aligned} \quad (3)$$

There are also four cross derivatives, at the corner points as

$$\begin{aligned} f_{uv}(0, 0) &= a_{11} \\ f_{uv}(1, 0) &= a_{11} + 2a_{21} + 3a_{31} \\ f_{uv}(0, 1) &= a_{11} + 2a_{12} + 3a_{13} \\ f_{uv}(1, 1) &= \sum_{i=1}^3 \sum_{j=1}^3 i j a_{ij} \end{aligned} \quad (4)$$

which in total provide the minimum information required to uniquely obtain the 16 unknown coefficients a_{ij} , $i, j = 0, 1, 2, 3$. Writing the above equations in the matrix form $y = Ax$, where A is the 16×16 design matrix and y is the observation vector, we may obtain the unknown vector as $x = A^{-1}y$. This procedure yields a surface patch on the unit square (0,0), (1,0), (0,1) and (1,1), which is continuous and has continuous the first and second derivatives.

In many practical applications, only the function values are available, but not the first and the cross derivatives. These derivatives can be obtained in different manners. They can for example be approximated by the finite difference method as follows [36,37]:

$$\begin{aligned} f_u(u, v) &= [f(u+1, v) - f(u-1, v)]/2 \\ f_v(u, v) &= [f(u, v+1) - f(u, v-1)]/2 \\ f_{uv}(u, v) &= [f(u+1, v+1) - f(u-1, v-1) + f(u, v)]/4 \end{aligned} \quad (5)$$

which uses the available function values at the adjacent points.

A drawback of the above interpolation problem is that it assumes the above derivatives are known at the corners. A better criterion, we will follow for LS-BICSA, is that the adjacent patches have continuous derivatives on the borders in general and at the corners in particular. For the pure interpolation problem, such constraints cannot introduce the minimum 16 equations required to solve the 16 unknown coefficients of each spline. This can be handled by LS-BICSA.

3. Least squares 2D bi-cubic spline approximation

The bi-cubic spline interpolation model has no redundancy and hence the unknown vector can uniquely be calculated. In this method, each interpolated surface passes through its control points at the four corners. We note however that such a perfect interpolation method is only useful if there is no error in the regular grid measured data. But, in practice, very often the measurements are contaminated with an unknown amount of noise. This will then indicate that ‘approximation’ is preferred over ‘interpolation’. An approximating polynomial does not necessarily pass through the defining knots (cf. Fig. 1); it is only required that the polynomial gets sufficiently close to its data points. An accurate way of approximation is based on the least squares principle, which is followed in this contribution.

3.1. Least squares method with hard constraints

We now introduce the least squares 2D bi-cubic spline approximation (LS-BICSA) method for the surface fitting problem. To this end, consider the following linear model of observation equations:

$$E(y) = Ax \text{ or } y = Ax + e; D(y) = Q_y \quad (6)$$

where A is the $m \times n$ design matrix, Q_y is the $m \times m$ covariance matrix of the observables y , which contains the variances on its diagonals and the covariances on its off-diagonals. E denotes the expectation operator and D stands for the dispersion operator. There are many possible ways to handle this problem of which the least squares is a standard method which minimizes $e^T Q_y^{-1} e = (y - Ax)^T Q_y^{-1} (y - Ax)$ [38]. This follows the weighted least squares estimate $\hat{x}$ as

$$\hat{x} = (A^T Q_y^{-1} A)^{-1} A^T Q_y^{-1} y \quad (7)$$

When the covariance matrix Q_y of observables is known, the above equation provides the best linear unbiased estimation (BLUE) of x [15]. Incomplete knowledge of the covariance matrix can occur in many applications. The estimation of unknown parameters of a covariance matrix is generally referred to as variance component estimation (VCE). We may at least refer to the Refs. [39,40]. The least-squares variance component estimator (LS-VCE) is one of the VCE methods applicable to many science and engineering applications [41–43].

The least squares estimate of the observations and residuals follows as $\hat{y} = A\hat{x} = P_A y$ and $\hat{e} = y - \hat{y} = P_A^\perp y$, respectively, where $P_A = A(A^T Q_y^{-1} A)^{-1} A^T Q_y^{-1}$ and $P_A^\perp = I_m - P_A$ are two orthogonal projectors [15] and I_m is an identity matrix of size m .

For the application of the least squares method to the LS-BICSA it is relevant to briefly review the theory of least squares subject to hard constraints. We closely follow Ref. [15]. Let us assume that we are given the linear model of observation equations in Eq. (6) subject to a few hard constraints on the parameters as

$$B^T x = 0 \quad (8)$$

where B is the $n \times b$ constraint matrix. The general solution of the homogeneous equation $B^T x = 0$, is given by

$$x = B^\perp \lambda \quad (9)$$

where λ is an $(n-b)$ -vector and $B^\perp$ is a $n \times (n-b)$ matrix of which its column vectors are orthogonal to B (i.e., $B^T B^\perp = 0$). Substituting the preceding equation into Eq. (9), yields

$$E(y) = AB^\perp \lambda; \quad D(y) = Q_y \quad (10)$$

The least squares estimate $\hat{\lambda}$ of λ is

$$\hat{\lambda} = (B^{\perp T} A^T Q_y^{-1} A B^\perp)^{-1} B^{\perp T} A^T Q_y^{-1} y \quad (11)$$

Substituting Eq. (11) into Eq. (9) gives the least squares estimate $\hat{x}$ of x as

$$\hat{x} = B^\perp \hat{\lambda} \quad (12)$$

For more information, the reader is referred to the book [15]. This theory is used in the least squares 2D bi-cubic spline approximation (LS-BICSA) of the next subsection.

The bi-cubic spline interpolation is used only for interpolating data points on a two-dimensional regular grid for which a set of grid points of a rectangular region is required. The LS-BICSA does not however necessarily require a set of regular grids. The specified patches can have in general any number of data points distributed (ir)regularly on the patches. Also the patches are not necessarily connected to each other on the data points; they can be connected in some arbitrary intermediate points, known as 'knots'. In a 2D bi-cubic spline interpolation we must have the function values and the first and cross derivatives available in the boundary knots. But in the LS-BICSA we do not require the function values and the first, second and cross derivatives at the knots to be known. For further clarification we propose two scenarios on the way the patches are connected to each other, which follow in the next subsection.

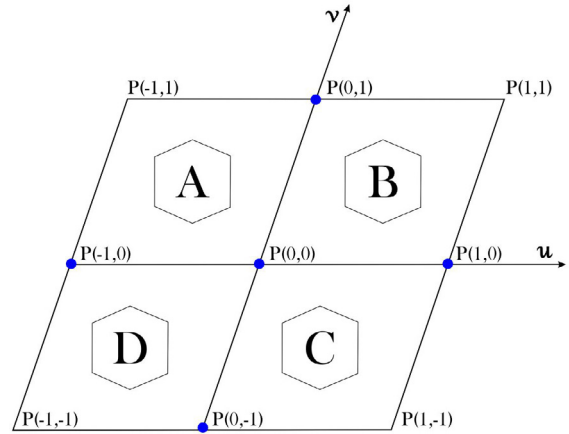


Fig. 3. Scenario I: Four surface patches with two boundary points along each side (blue knots) to be connected to each other along abscissa and ordinate. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.2. Application to LS-BICSA

We now apply the theory of the previous subsection to the problem of 2D bi-cubic spline approximation. Hard constraints, in the form of Eq. (8), are introduced to the fitted spline or its derivatives to fulfill the continuity conditions at the adjacent points. Although the constraints can be introduced in different forms, we propose only the following two scenarios. The continuity properties of these two scenarios are then explained.

Scenario I: Suppose we are given a set of m known data points (u_j, v_j) , $j = 1, \dots, m$ possibly distributed irregularly on a plane surface over a rectangular region $R: [-1 < u < 1; -1 < v < 1]$. The $w_j = \phi(u_j, v_j)$ of the above points has been measured, as a function value. In this surface fitting problem, u and v are the independent variables and w is the dependent one; the w values are assumed to be observed on the known points. The aim is to create a surface $\phi(u, v)$ which best approximates the given data set using the least squares criterion. We divide the region into four patches, A, B, C and D—to give an example (Fig. 3). The patches are connected to each other longitudinally (along u -axis) and laterally (along v -axis) and each has m_i , $i = 1, \dots, 4$ data points, and hence m_i , $i = 1, \dots, 4$ measured values of w such that $m = \sum_{i=1}^4 m_i$ becomes the total number of measurements. In scenario I, these patches are connected together by only two knots at their boundaries; we thus impose the continuity constraints at these boundary points.

The bi-cubic spline, made of four surface patches in Fig. 3, is of the form

$$\phi(u, v) = \begin{cases} \phi^{(A)}(u, v) = \sum_{i=0}^3 \sum_{j=0}^3 a_{ij} u^i v^j; u \in [-1, 0], v \in [0, +1] \\ \phi^{(B)}(u, v) = \sum_{i=0}^3 \sum_{j=0}^3 b_{ij} u^i v^j; u \in [0, +1], v \in [0, +1] \\ \phi^{(C)}(u, v) = \sum_{i=0}^3 \sum_{j=0}^3 c_{ij} u^i v^j; u \in [0, +1], v \in [-1, 0] \\ \phi^{(D)}(u, v) = \sum_{i=0}^3 \sum_{j=0}^3 d_{ij} u^i v^j; u \in [-1, 0], v \in [-1, 0] \end{cases} \quad (13)$$

which are to be fitted to the observations at all $m = \sum_{i=1}^4 m_i$ data points of the four patches in the least squares sense. The coefficients $(a_{ij}, b_{ij}, c_{ij}, d_{ij})$; $i, j = 0, 1, 2, 3$ are the unknown parameters to be estimated. Given m known points, and hence m observed values, there is in total m observation equations and $4 \times 16 = 64$ unknown coefficients.

The fitted splines and their derivatives may take specified values at their boundary knots. It is however not necessary to know these values. We just need to impose some continuity and smoothness conditions. For our application, it is enough that the function values ϕ , its first order

derivatives ϕ_u , ϕ_v , its cross derivatives ϕ_{uv} and its second order derivatives ϕ_{uu} and are identical for both functions on either side of the boundary knots. For the longitudinal (u-axis) connection of patches A and B, we just need 10 independent continuity conditions at boundary points (0,0) and (0,1). They are as follows

$$\begin{aligned}\phi^{(A)}(u_{BP}, v_{BP}) &= \phi^{(B)}(u_{BP}, v_{BP}) \\ \phi_u^{(A)}(u_{BP}, v_{BP}) &= \phi_u^{(B)}(u_{BP}, v_{BP}) \\ \phi_v^{(A)}(u_{BP}, v_{BP}) &= \phi_v^{(B)}(u_{BP}, v_{BP}) \\ \phi_{uu}^{(A)}(u_{BP}, v_{BP}) &= \phi_{uu}^{(B)}(u_{BP}, v_{BP}) \\ \phi_{uv}^{(A)}(u_{BP}, v_{BP}) &= \phi_{uv}^{(B)}(u_{BP}, v_{BP})\end{aligned}\quad (14)$$

where the boundary points are $(u_{BP}, v_{BP}) = (0, 0)$ and $(u_{BP}, v_{BP}) = (0, 1)$. Two observations are highlighted as follows. The first observation is that the constraints $\phi_{uv}^{(A)}(u_{BP}, v_{BP}) = \phi_{uv}^{(B)}(u_{BP}, v_{BP})$ are not independent from those provided in Eq. (14), and hence they do not provide new information. The second observation is that the above constraints at knots (0,0) and (0,1) makes the two patches continuous and differentiable along the u and v axes. This also holds for the ϕ_{uu} , which is a continuous function. We however note that ϕ_{uv} is not continuous on the intermediate points between (0,0) and (0,1).

In a similar manner, for patches B and C, 10 continuity conditions are required to join the fitted surfaces at boundary knots (0,0) and (1,0). Because these two patches are laterally connected to each other (v axis), the constraints are of the form

$$\begin{aligned}\phi^{(B)}(u_{BP}, v_{BP}) &= \phi^{(C)}(u_{BP}, v_{BP}) \\ \phi_u^{(B)}(u_{BP}, v_{BP}) &= \phi_u^{(C)}(u_{BP}, v_{BP}) \\ \phi_v^{(B)}(u_{BP}, v_{BP}) &= \phi_v^{(C)}(u_{BP}, v_{BP}) \\ \phi_{vv}^{(B)}(u_{BP}, v_{BP}) &= \phi_{vv}^{(C)}(u_{BP}, v_{BP}) \\ \phi_{uv}^{(B)}(u_{BP}, v_{BP}) &= \phi_{uv}^{(C)}(u_{BP}, v_{BP})\end{aligned}\quad (15)$$

where the boundary points are $(u_{BP}, v_{BP}) = (0, 0)$ and $(u_{BP}, v_{BP}) = (1, 0)$.

The last patch, D, should be connected to both patches A and C along the u and v axes. First, assume that the patch D is connected to patch C. We then have again 10 continuity conditions at boundary knots (0,0) and (0, -1), just like patches A and B. Then the patch D is to be connected laterally to the patch A. One may think that again 10 continuity conditions are required. This however is not the case. This time are only four constraints (out of the ten) required at the boundary point $(-1, 0)$ between A and D. This is because the patch D, through C and B, has been already connected the patch A. This indicates that only four independent constraints are required to connect D to A, and hence the additional continuity conditions create dependent constraints. Table 1 summarizes the required constraints at the boundary points for scenario I. Therefore there exist 34 hard constraints, to be added to the system of equations. The redundancy of the model is then $df = m - (4 \times 16) + 34$; we thus need at least $m = 30$ observations to

uniquely solve this model.

To solve these observation equations with the hard constraints using the least squares estimate $\hat{x}$ in Eq. (12), we require the design matrix A and the constraint matrix B. The design matrix has a block-diagonal structure as $A = \text{blkdiag}([A_1, A_2, A_3, A_4])$, where blkdiag constructs a block-diagonal matrix from its input arguments, and A_i , $i = 1, \dots, 4$ is formed for each of the patches A, B, C or D (see Appendix A.1).

In addition, the unknown vector x and the m -vector of observations y given in Eq. (6) have the following structure:

$$\begin{aligned}x_{64 \times 1} &= [a_{ij} b_{ij} c_{ij} d_{ij}]^T; i, j = 0, 1, 2, 3 \\ y_{m \times 1} &= [\phi_{1 \times m_1}^{(A)} \phi_{1 \times m_2}^{(B)} \phi_{1 \times m_3}^{(C)} \phi_{1 \times m_4}^{(D)}]^T\end{aligned}\quad (16)$$

Accordingly, the 64×34 constraint matrix B, in Eq. (8), is of the form

$$B^T = \begin{bmatrix} B_1^{(0,0)} & -B_1^{(0,0)} & Z & Z \\ B_1^{(0,1)} & -B_1^{(0,1)} & Z & Z \\ Z & B_2^{(0,0)} & -B_2^{(0,0)} & Z \\ Z & B_2^{(1,0)} & -B_2^{(1,0)} & Z \\ Z & Z & B_1^{(0,0)} & -B_1^{(0,0)} \\ Z & Z & B_1^{(0,-1)} & -B_1^{(0,-1)} \\ B_3^{(-1,0)} & Z & Z & -B_3^{(-1,0)} \end{bmatrix}\quad (17)$$

where the submatrices $B_1^{(u,v)}$, $B_2^{(u,v)}$ and $B_3^{(u,v)}$ are presented in Appendix A.2. Blocks Z are zero matrices with an appropriate size. We then can compute the unknown coefficients by substituting the observation vector y , the design matrix A and the constraint matrix B into Eqs. (11) and (12). The least squares estimate of the residuals (estimated errors) is $\hat{e} = y - A\hat{x} = y - AB^T \hat{\lambda}$.

Scenario II: In this scenario we again assume that we are given a set of m known knots (u_j, v_j, w_j) , $j = 1, \dots, m$ distributed randomly on an unknown surface over a given region R: $[-1 < u < 1; -1 < v < 1]$.

Without loss of the generality, we again divide the region into four patches A, B, C and D. But in this scenario, these patches are connected to each other at four knots (two points at the corners and two intermediate points). The continuity and smoothness conditions are thus imposed at these four boundary points (see Fig. 4).

The bi-cubic spline equations are exactly the same as those given in Eq. (13). They are to be fitted to m_i , $i = 1, \dots, 4$ measured values at the known points of four patches. In a similar manner, the coefficients $(a_{ij}, b_{ij}, c_{ij}, d_{ij})$, $i, j = 0, 1, 2, 3$ are the unknown parameters to be estimated. But for the longitudinal (along u-axis) connection of patches A and B, in this case, we have indeed 12 continuity conditions at boundary points $BP_{AB} = \{(0, 0), (0, 1/3), (0, 2/3), (0, 1)\}$. They are given as

$$\begin{aligned}\phi^{(A)}(u_{BP}, v_{BP}) &= \phi^{(B)}(u_{BP}, v_{BP}) \\ \phi_u^{(A)}(u_{BP}, v_{BP}) &= \phi_u^{(B)}(u_{BP}, v_{BP}) \\ \phi_{uu}^{(A)}(u_{BP}, v_{BP}) &= \phi_{uu}^{(B)}(u_{BP}, v_{BP})\end{aligned}\quad (18)$$

Table 1

Specified independent constraints in scenario I; : indicates that two patches are connected at two boundary points.

Type of Constraints	Number of Constraints/Points used							
	A : B	Boundary Points	B : C	Boundary Points	D : C	Boundary Points	A : D	Boundary Points
$\phi(u_{BP}, v_{BP})$	2	(0,0),(0,1)	2	(0,0), (1,0)	2	(0,0), (0, -1)	1	(-1,0)
$\phi_u(u_{BP}, v_{BP})$	2	(0,0),(0,1)	2	(0,0), (1,0)	2	(0,0), (0, -1)	0	-
$\phi_v(u_{BP}, v_{BP})$	2	(0,0),(0,1)	2	(0,0), (1,0)	2	(0,0), (0, -1)	1	(-1,0)
$\phi_{uu}(u_{BP}, v_{BP})$	2	(0,0),(0,1)	0	-	2	(0,0), (0, -1)	0	-
$\phi_{vv}(u_{BP}, v_{BP})$	0	-	2	(0,0), (1,0)	0	-	1	(-1,0)
$\phi_{uv}(u_{BP}, v_{BP})$	2	(0,0),(0,1)	2	(0,0), (1,0)	2	(0,0), (0, -1)	1	(-1,0)
SUM	10		10		10		4	

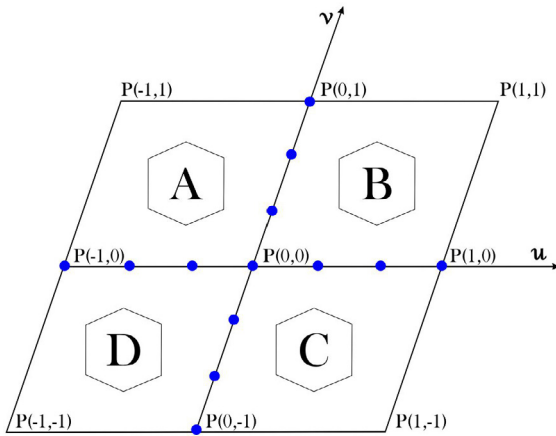


Fig. 4. Scenario II: Four surface patches with four boundary points along each side (blue knots) are connected to each other along abscissa and ordinate. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

where $u_{BP} = 0, v_{BP} = 0, 1/3, 2/3, 1$. Note that the intermediate knots $1/3$ and $2/3$ are just an example because theoretically they may take any value between 0 and 1. The conditions $\phi_v^{(A)} = \phi_v^{(B)}$ and $\phi_{vv}^{(A)} = \phi_{vv}^{(B)}$ introduce dependent constraints and hence they do not provide new information. It is because the patches A and B have been connected to each other through four points along the v -axis and hence a common 1D cubic polynomial (as a function of v) is the intersection of patches A and B. Therefore all higher derivatives are automatically continuous along the v -axis.

For patches B and C, again 12 continuity conditions are imposed to join the fitted surface patches at boundary points $BP_{BC} = \{(0, 0), (1/3, 0), (-2/3, 0), (1, 0)\}$

$$\begin{aligned}\phi^{(B)}(u_{BP}, v_{BP}) &= \phi^{(C)}(u_{BP}, v_{BP}) \\ \phi_v^{(B)}(u_{BP}, v_{BP}) &= \phi_v^{(C)}(u_{BP}, v_{BP}) \\ \phi_{vv}^{(B)}(u_{BP}, v_{BP}) &= \phi_{vv}^{(C)}(u_{BP}, v_{BP})\end{aligned}\quad (19)$$

where $u_{BP} = 0, 1/3, 2/3, 1, v_{BP} = 0$. The last patch, D, should be connected to both patches A and C along the u and v axes. First, assume that the patch D is connected to patch C. We then have again 12 continuity conditions at boundary points $BP_{CD} = \{(0, 0), (0, -1/3), (0, -2/3), (0, -1)\}$, just like patches A and B. Then the patch D must be connected laterally to the patch A. But this time are only three (out of the 12) constraints required at only one of the boundary points BP_{DA} is either of $\{(-1/3, 0), (-2/3, 0), (-1, 0)\}$. Table 2 shows the required number of constraints at the boundary points for scenario II. Therefore, there exist in total 39 hard constraints, to be added to the system of equations. The redundancy of this model is then $df = m - (4 \times 16) + 39$, and hence one requires at least 25 observations to uniquely solve this model.

The design matrix A has also the same block-diagonal form

Table 2

Specified independent constraints in scenario II; :: indicates that two patches are connected at four boundary points.

Type of Constraints	Number of Constraints/Points used							
	A::B	Boundary Points	B::C	Boundary Points	D::C	Boundary Points	A::D	Boundary Points
$\phi(u_{BP}, v_{BP})$	4	BP_{AB}	4	BP_{BC}	4	BP_{CD}	1	BP_{DA}
$\phi_u(u_{BP}, v_{BP})$	4	BP_{AB}	0	–	4	BP_{CD}	0	–
$\phi_v(u_{BP}, v_{BP})$	0	–	4	BP_{BC}	0	–	1	BP_{DA}
$\phi_{uu}(u_{BP}, v_{BP})$	4	BP_{AB}	0	–	4	BP_{CD}	0	–
$\phi_{vv}(u_{BP}, v_{BP})$	0	–	4	BP_{BC}	0	–	1	BP_{DA}
SUM	12		12		12		3	

$A = blkdiag([A_1, A_2, A_3, A_4])$ as that of scenario I (see Appendix A.1). In addition, the unknown vector x and the m -vector of observations y have the same structure as those provided in Eq. (16). The constraint matrix $B_{(4 \times 16) \times 39}$ is however different. It is of the form

$$B^T = \begin{bmatrix} B_1^{(0,0)} & -B_1^{(0,0)} & Z & Z \\ B_1^{(0,1/3)} & -B_1^{(0,1/3)} & Z & Z \\ B_1^{(0,2/3)} & -B_1^{(0,2/3)} & Z & Z \\ B_1^{(0,1)} & -B_1^{(0,1)} & Z & Z \\ Z & B_2^{(0,0)} & -B_2^{(0,0)} & Z \\ Z & B_2^{(1/3,0)} & -B_2^{(1/3,0)} & Z \\ Z & B_2^{(2/3,0)} & -B_2^{(2/3,0)} & Z \\ Z & B_2^{(1,0)} & -B_2^{(1,0)} & Z \\ Z & Z & B_1^{(0,0)} & -B_1^{(0,0)} \\ Z & Z & B_1^{(0,-1/3)} & -B_1^{(0,-1/3)} \\ Z & Z & B_1^{(0,-2/3)} & -B_1^{(0,-2/3)} \\ Z & Z & B_1^{(0,-1)} & -B_1^{(0,-1)} \\ B_3^{(BP_{DA})} & Z & Z & -B_3^{(BP_{DA})} \end{bmatrix} \quad (20)$$

where the submatrices, $B_1^{(u,v)}$, $B_2^{(u,v)}$ and $B_3^{(u,v)}$ are presented in Appendix A.3. The blocks Z are some zero matrices with an appropriate size. The above given matrices can be employed to compute the unknown coefficients using Eqs. (11) and (12).

A final remark on the continuity properties of Scenarios I and II is in order. When considering Figs. 3 and 4, it follows that Scenario I imposes 34 continuity and differentiability constraints, whereas Scenario II imposes 39 constraints. This indicates that Scenario I has lower redundancy and hence looser continuity and differentiability constraints when compared with Scenario II. Our observations indicate, for Scenario I, that the patches are continuous and differentiable along the u and v axes. This scenario, however, does not impose the second differentiability condition across the borders. This is not the case for Scenario II, which has all of the continuity and differentiability constraints up and including to the second order along and across the borders. Employing all 39 constraints of Scenario II, the method proposed by [11] is considered to be a special case of this scenario.

4. Numerical results and discussions

In this section, the efficiency of the proposed surface fitting problem using LS-BICSA is presented through two numerical examples. The employment of the proposed method for approximating real data is illustrated. The first application is a mathematical function approximation whereas the second example is a real geoid height approximation. Our expectation in both examples is to achieve user-defined error values between the actual and approximated function values.

Example 1

Consider the following mathematical function

$$f(u, v) = ue^{(-u^2 - v^2)} \quad (21)$$

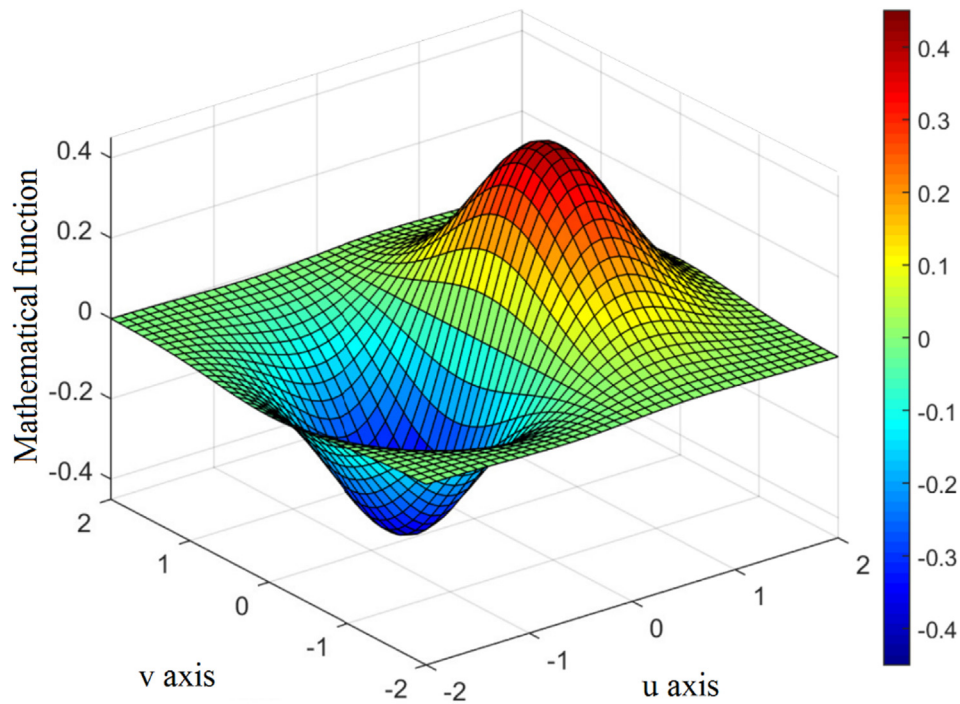


Fig. 5. Mathematical function $f(u, v) = ue^{(-u^2-v^2)}$ illustrated on a rectangular regular grid of $u = -2: 0.1: 2$ and $v = -2: 0.1: 2$.

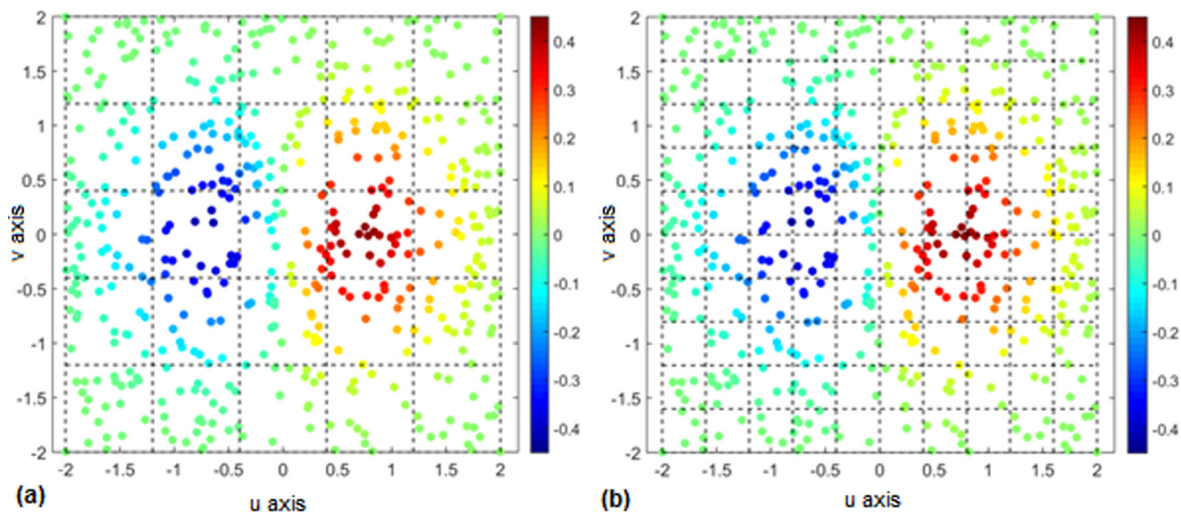


Fig. 6. 500 data points generated randomly for mathematical function $f(u, v) = ue^{(-u^2-v^2)}$ on an irregular grid over region $-2 \leq u, v \leq 2$; a (Case 3 with 25 patches), b (Case 5 with 100 patches).

Fig. 5 shows the above function on a regular grid of $41 \times 41 = 1681$ points on $u = -2: 0.1: 2$ and $v = -2: 0.1: 2$. A data set of 500 points, generated using the uniform distribution over the region $-2 \leq u, v \leq 2$, is used to illustrate the efficacy of the proposed method (Fig. 6); the uniform distribution generates each member of the family in a manner that all intervals of the same length are equally probable. The goal is to approximate this data set using the 2D bi-cubic splines theory proposed in Section 3. In this simulation study, the covariance matrix Q_y is assumed to be an identity matrix. Five cases are investigated for the surface patch sizes ranging from 2×2 to 0.4×0.4 . These five cases correspond to 4, 16, 25, 64, and 100 surface patches. The results for Cases 3 and 5 are investigated in more details. Case 3) The entire 4×4 region is divided into 25 surface patches of size 0.8×0.8 (Fig. 6a), Case 5) The entire region is divided into 100 surface patches of size 0.4×0.4 (Fig. 6b). The results are only presented for scenario II. Similar results can be obtained for scenario I.

LS-BICSA is applied to the irregular data set in Fig. 6 to estimate the unknown coefficients of the surface patches. The estimated coefficients of the approximated surface patches are then used to obtain the function values $f(u, v)$ on a regular grid $u = -2: 0.1: 2$ and $v = -2: 0.1: 2$ with 1681 points. In fact, the estimated 2D bi-cubic polynomials approximate the actual mathematical function $f(u, v)$ in Fig. 5. Table 3 provides some statistics of the above five cases. For each case, it includes the number of patches, number of observations, number of spline parameters, number of continuity constraints, number of unknown coefficients and the degree of freedom. The number of unknown coefficients are the length of the vector $\hat{\lambda}$ in Eq. (11) and the degree of freedom will be $df = m - (n - b)$. The maximum differences between the actual and approximated function values are also presented in Table 3. The RMSE values of the differences are also provided in this table. For further clarification we explain Cases 3 and 5 in more details.

Case 3: In this case we divide the entire study region into 25 patches

Table 3

Statistics of different cases and their estimated errors in scenario II for mathematical function $f(u, v) = ue^{(-u^2-v^2)}$ approximated on a rectangular regular grid of $u = -2: 0.1: 2$ and $v = -2: 0.1: 2$.

Case	Patch size	No. of Patches	No. of Obs.	No. of SplinePar.	No. of Const.	No. of Unkn.	Degree of Free.	MAX error	RMSE	MeanRMSE (100 data)
1	2×2	4	500	64	39	25	475	0.189	0.058	0.059
2	1×1	16	500	256	207	49	451	-0.070	0.021	0.022
3	0.8×0.8	25	500	400	336	64	436	-0.016	0.004	0.004
4	0.5×0.5	64	500	1024	903	121	379	4.2×10^{-3}	8×10^{-4}	9×10^{-4}
5	0.4×0.4	100	500	1600	1431	169	331	1.3×10^{-3}	3×10^{-4}	5×10^{-4}

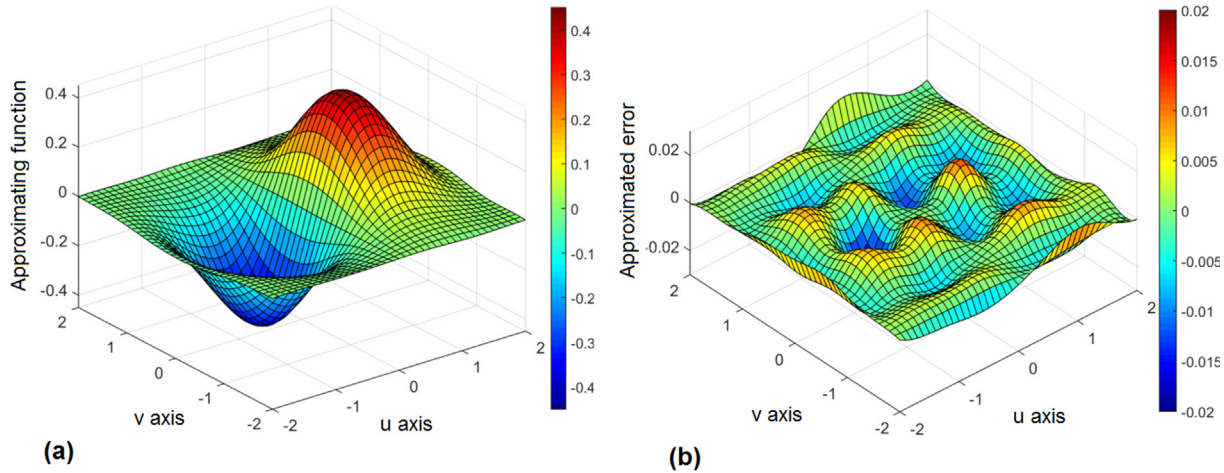


Fig. 7. Mathematical function $f(u, v) = ue^{(-u^2-v^2)}$ approximated using 2D bi-cubic splines of Case 3 and irregular grid data. Approximating function on a regular grid of $u = -2: 0.1: 2$ and $v = -2: 0.1: 2$ (a), the difference between the actual and approximated function values (approximating error) (b).

with the size of 0.8×0.8 as shown in Fig. 6a. We have in total 25 patches, $25 \times 16 = 400$ spline parameters and the total number of observations within the entire region is 500. There are a few known points of the function values randomly distributed within each surface patch. We impose 336 constraints at the four boundary knots under scenario II. The redundancy of the model is then 436. Fig. 7a shows the approximated function $f(u, v)$ using the least squares 2D bi-cubic spline approximation method on a regular grid. The difference between the actual and approximated function values (error values) is illustrated in Fig. 7b. The results indicate that the maximum error and the RMSE values are equal to 16×10^{-3} and 4×10^{-3} , respectively.

Case 5: In this case we divide the entire study region into 100 patches with the size of 0.4×0.4 as shown in Fig. 6b. We have in total 100 patches and $100 \times 16 = 1600$ spline parameters. The total number of observations within the entire region is again 500. There are a few known points of the function values randomly distributed within each surface patch. We impose 1431 constraints at the four boundary knots. The redundancy is then 331. Fig. 8a shows the approximated function $f(u, v)$ using the LS-BICSA method on a regular grid. The difference between the actual and approximated function values (error values) is illustrated in Fig. 8b. The results indicate that the maximum error and the RMSE values are equal to 1.3×10^{-3} and 3×10^{-4} , respectively.

Figs. 7 and 8 illustrate that the approximation accuracy is improved by decreasing the size of the patches. In other words, decreasing the patch size can provide smaller approximating errors and hence a better approximation of the actual function. In principle, to achieve a finer resolution, the patch sizes should get smaller and smaller. We can reduce the patch size while the number of unknown parameters in Eq. (10) remains smaller than the number of observations (more rows than columns in AB^T). This is however a necessary condition. A sufficient condition may also require that the design matrix AB^T is still of full column rank.

To further verify the efficacy of the proposed method, for the above

five cases, the simulation procedure was repeated over 100 different runs. For each run, the sample set was produced again using the uniform distribution of 500 generated points over the 2D region. The mean RMSE values (over 100 runs) of the differences between the actual and approximated function values are also given in the last column of Table 3. Further detailed information on the results of 100 different simulation data is also provided in this table. The results confirm that the approximations are adequately accurate and hence indicating the reliability and stability of the proposed method.

It is highlighted that LS-BICSA has been implemented to a series of other mathematical functions. The results are comparable to what we obtained in Example 1. For the sake of brevity, we avoid to repeat this procedure for other mathematical functions in the present contribution. Instead, we employ a real data set to further elaborate on the efficiency of the proposed method in the next example.

Example 2

The second data set is the real geoid height obtained from a Global Geopotential Model (GGM). Geoid is an equipotential surface that best approximates the global mean sea level. A GGM contains a set of spherical harmonic coefficients for the earth's gravity potential. These coefficients are derived from satellite measurements, terrestrial and altimetry gravity data. We may express the geoid height as the series of these coefficients at any point of interest (φ, λ) with respect to the reference ellipsoid [44].

The gravity field model EIGEN-6C4 and the reference ellipsoid WGS84 are selected in this paper. WGS84 is an earth-centered, earth-fixed reference ellipsoid defined by the United States National Geospatial-Intelligence Agency (NGA). EIGEN-6C4 is a combined global gravity field model that includes GOCE, GRACE, LAGEOS and ground and altimetry gravity data up to degree and order 2190 [45]. It can be obtained from the ICGEM data base at GFZ Potsdam via <http://icgem.gfz-potsdam.de> [46]. Because the standard deviation of the spherical harmonic coefficients is given by EIGEN-6C4, the covariance matrix of

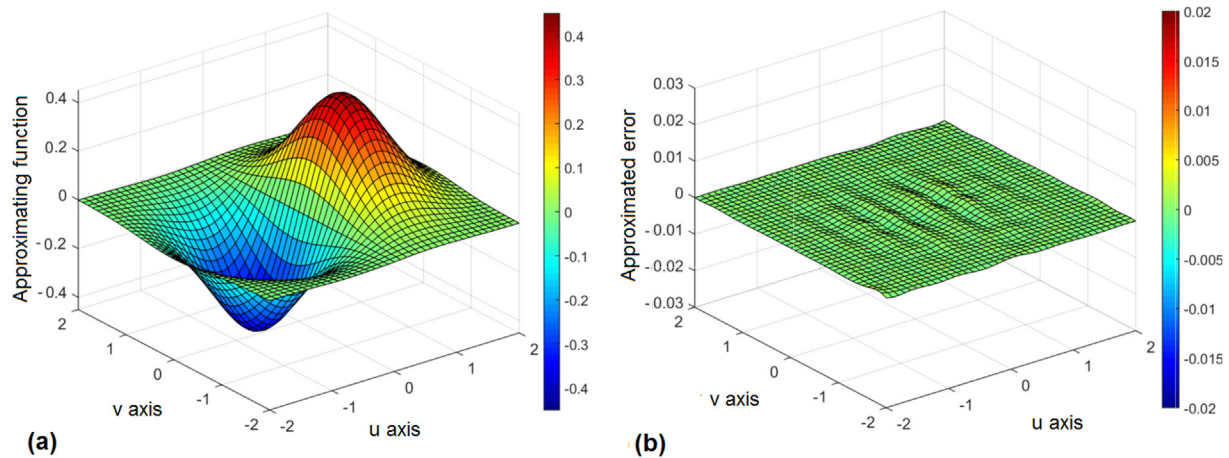


Fig. 8. Mathematical function $f(u, v) = ue^{-(u^2-v^2)}$ approximated using 2D bi-cubic splines of Case 5 and irregular grid data. Approximating function on a regular grid of $u = -2: 0.1: 2$ and $v = -2: 0.1: 2$ (a), the difference between the actual and approximated function values (approximating error) (b); vertical scale is the same as that in Fig. 7 b to make comparison easier.

geoid heights can be calculated through the error propagation law [38]. We thus use this covariance matrix in the least squares process of Eq. (11).

The geoid height data in an arbitrary area bounded by the latitude $[53^\circ\text{N} - 54^\circ\text{N}]$ and the longitude $[1^\circ\text{E} - 2^\circ\text{E}]$ was used in this study. The geoid heights were computed in terms of a high-resolution 1-arcmin grid covering this area. We then have 60×60 known geoid heights at the points $(\varphi_j, \lambda_j, N_j)$, $j = 1, \dots, 3600$ distributed on an unknown surface over a rectangular 1-degree region. The goal is to create a surface which approximates the given data set by the proposed LS-BICSA method. To illustrate the efficacy of the proposed method, 1296 points were selected as check points and hence excluded from the analysis (all data on the $5' \times 5'$ grid). LS-BICSA is thus applied to 2304 remaining data to estimate the unknown coefficients of the surface patches. The estimated coefficients are then used to obtain the function values on the excluded check points.

Fig. 9 shows the study area along with its geoid heights on 1-arcmin spacing grids with 3600 data points. The region is investigated under

five different cases with the size patches of $30' \times 30'$, $20' \times 20'$, $15' \times 15'$, $10' \times 10'$ and $5' \times 5'$. The surface is thus approximated for the above five cases. The approximation errors between the actual and approximated function values are calculated at the check points for the specified patches of different sizes. The maximum errors and the RMSE values of the differences are also provided in each case.

Each of the above five cases are investigated in two scenarios described in the previous section, i.e. scenario I in which patches are connected together at only two corner points, and scenario II in which patches are connected together at four boundary points. Here we describe two extreme cases of $30' \times 30'$ patch size (Case 1) and of $5' \times 5'$ patch size (Case 2), in two scenarios. The detailed information is provided in Tables 4 and 5.

Case 1: In this case we divide the study region into patches with the size of $30' \times 30'$. Each patch has 576 data points. We thus have in total four patches, $4 \times 16 = 64$ spline parameters and 2304 observations. 1296 points are also considered as check points. Considering scenario I, we require 34 constraints to connect these patches at the boundary

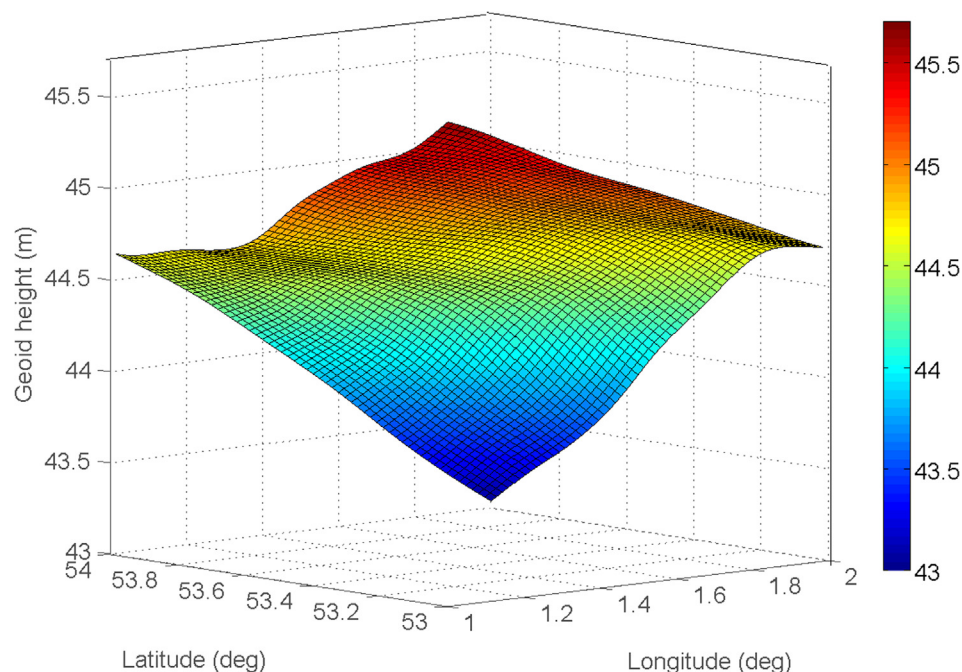


Fig. 9. Geoid heights (in meter) of $1^\circ \times 1^\circ$ study area on 1-arcmin grid data (3600 data points).

Table 4
Statistics of different cases and their estimated residual vectors in scenario I.

Case	Patch size	No. of Patches	No. of Obs.	No. of Spline Par.	No. of Const.	Degree of Free.	MAX Err. [mm]	RMSE [mm]
1	30' × 30'	2 × 2	2304	64	34	2274	-42.5	11.2
2	20' × 20'	3 × 3	2304	144	96	2256	25.5	7.6
3	15' × 15'	4 × 4	2304	256	186	2234	-23.1	5.7
4	10' × 10'	6 × 6	2304	576	450	2178	12.6	3.4
5	5' × 5'	12 × 12	2304	2304	1914	1914	1.7	0.4

Table 5
Statistics of different cases and their estimated residual vectors in scenario II.

Case	Patch size	No. of Patches	No. of Obs.	No. of Spline Par.	No. of Const.	Degree of Free.	MAX Err. [mm]	RMSE [mm]
1	30' × 30'	2 × 2	2304	64	39	2279	-45.8	12.9
2	20' × 20'	3 × 3	2304	144	108	2268	27.4	10
3	15' × 15'	4 × 4	2304	256	207	2255	-22.8	7.9
4	10' × 10'	6 × 6	2304	576	495	2223	-12.3	4.3
5	5' × 5'	12 × 12	2304	2304	2079	2079	1.6	0.4

knots. The degree of freedom of the model is then $df = 2304 - 64 + 34 = 2274$. The difference between the actual and approximated function values at the check points (error values) is calculated. The results indicate that the maximum error and the RMSE values are equal to -42.5 mm and 11.2 mm, respectively. For scenario II, we have 64 unknown parameters, 2304 observations, 39 boundary conditions and hence 2279 degrees of freedom. The results indicate that the maximum error and the RMSE values are equal to -45.8 mm and 12.9 mm, respectively. It is noted that the range of variations of the error values for scenario I is slightly smaller than those for scenario II. This makes sense because scenario I imposes smaller number of constraints to the functional model. The smaller the number of constraints is, the smaller the redundancy of the model is, and hence the looser the functional model will be. Fig. 10a shows the estimated geoid height using LS-BICSA at 2304 data points. The difference between the actual and approximated function values at the check points (error values) is illustrated in Fig. 10b. As observed, the approximation error is below 5 cm.

Case 5: In this case, we divide the entire study region into patches with the size of 5' × 5'. Each surface patch has 16 known points. We thus have in total 144 patches, $144 \times 16 = 2304$ spline parameters and 2304 observations. 1296 points are also considered as check points. For scenario I, we impose 1914 constraints at the boundary knots. The redundancy of the model is then $df = 2304 - 2304 + 1914 = 1914$. The

difference between the actual and approximated function values at the check points (error values) is calculated. The results indicate that the maximum error and the RMSE values are equal to 1.7 mm and 0.4 mm, respectively. Therefore the error values, in this case, are significantly smaller compared to Case 1 with the patch size of 30' × 30'. Under scenario II, only does the number of constraints vary from scenario I. Because the patches are connected together at four boundary knots in this scenario, we have 2079 boundary constraints resulting in 2079 degrees of freedom. The results indicate that the maximum error and the RMSE values are equal to 1.6 mm and 0.4 mm, respectively (Table 5). The results are comparable to scenario I. Fig. 11a shows the geoid height approximated by LS-BICSA at 2304 data points. The difference between the actual and approximated function values at the check points (error values) is illustrated in Fig. 11b. The same scale bar as that presented in Fig. 10b was used to make the comparison easier. A clear reduction in the approximation error is observed.

The statistics for all Cases 1, 2, 3, 4 and 5, under scenario I and II, are given in Tables 4 and 5, respectively. The results show that the approximation accuracy is reduced by increasing the patch size in both scenarios. Decreasing the patch size indicates introducing more spline parameters, more continuity constraints, and hence more unknown parameters to be estimated. More unknown parameters avoid the problem of 'under parameterization'. In other words, decreasing the patch size can provide smaller approximation errors and hence better approximating function.

The tables also show the maximum approximation error along with the RMSE for the five cases mentioned above. These are general indications on the accuracy of the bi-cubic spline approximation. The results indicate that increasing the patch size degrades the approximation performance of the proposed method in both scenarios. Comparison between scenarios I and II (compare Tables 4 and 5) exhibits a slightly larger errors of scenario II compared to scenario I, especially in large-size patches. These issues make sense because when comparing scenarios I and II, by increasing the number of constraints in scenario II, the number of spline parameters remains constant. Therefore, the degree of freedom in scenario II increases and higher redundancy, due to fewer number of unknowns, will behave as 'under parameterization'. This, in general, can invalidate the estimates. Although Scenario 2 gives slightly larger errors, it provides more realistic results, since the surfaces have to be smooth at every boundary point.

5. Concluding remarks

We outlined the principle of employing the least squares 2D bi-cubic spline approximation (LS-BICSA) method to estimate a smoothed surface of a 2D data set. LS-BICSA, generally applicable to a set of irregular data, is formulated by the pure bi-cubic spline functions. The region

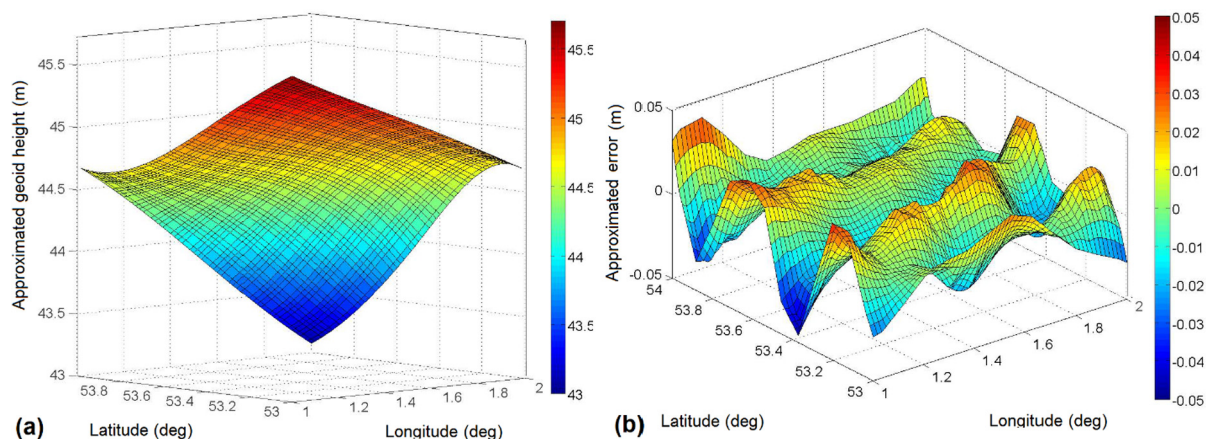


Fig. 10. Approximated geoid heights using LS-BICSA (a) along with its approximation error values (b) in $1^\circ \times 1^\circ$ study area, with 30' × 30' patch size in scenario II.

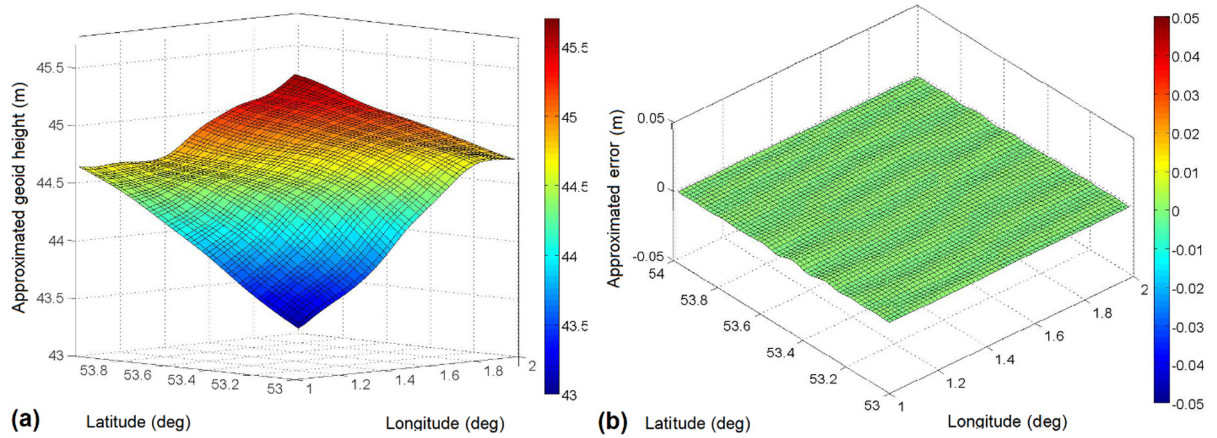


Fig. 11. Approximated geoid heights using LS-BICSA (a) along with its approximation error values (b) in $1^\circ \times 1^\circ$ study area, with $5' \times 5'$ patch size in scenario II.

must be divided into some patches and the patches must be connected together. This method does not necessarily require a set of grid knots or a rectangular network. The patches can have in general different number of known knots distributed randomly on the patches (Fig. 6). Also in this method we do not necessarily need the function values and the first and second derivatives to be known at the boundary of the patches. Only the continuity conditions are imposed at the boundary points of the patches to join together longitudinally and laterally. We then require to solve a least squares problem with hard constraints. These constraints were described in detail in two different scenarios. In scenario I, the patches were connected together by only two points at their boundaries, but in scenario II the patches were connected together by four points at their boundary knots.

To investigate the performance of the method, the results of two examples were presented. The first example is a known mathematical function, which was approximated using LS-BICSA on irregular 2D points. The second example was a real data set of the geoid height. The data was obtained from the gravity field model EIGEN-6C4 with reference to the WGS84 ellipsoid in an arbitrary small area. Geoid heights were computed in terms of a 1-arcmin grid covering the 1-degree study area. The aim was to create a surface which approximates the given data set by the proposed method. We divided the region into five different size patches. The residuals of the approximated surface were then calculated with a desired patch size under two scenarios. The results

indicated that the proposed method is capable of approximating the unknown surface from arbitrary data with (small) specified residuals. We observed that the approximation accuracy was reduced by increasing the patch size in both scenarios I and II. In other words, decreasing the patch size can provide smaller residuals and hence a better approximation. Comparison between scenarios I and II exhibited a slightly larger errors of the scenario II compared to the scenario I, especially in large-size patches. This method can thus be introduced as a reliable method for any application where one requires to find a smoothed 2D approximation of an irregularly distributed data set with an arbitrary accuracy.

Combination of different global and regional geoid models in a small-scale region has been the subject of intensive research in the past. In Ref. [47] this was achieved by employing 2D polynomials and the theory of LS-VCE. This can now be implemented to large-scale regions using the proposed LS-BICSA theory, which is to be adopted to LS-VCE. This is the subject of further research for future publications.

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Appendix

To solve the observation equations with the hard constraints using the least squares estimate $\hat{x}$ in Eq. (12), we require the design matrix A and the constraint matrix B . The design matrix A and the submatrices of the constraint matrix B in Eqs. (17), (20) are presented here.

A.1. Submatrices of design matrix A

The design matrix has a block-diagonal structure as $A = \text{blkdiag}([A_1, A_2, A_3, A_4])$, where blkdiag constructs a block-diagonal matrix from its input arguments, and A_i , $i = 1, \dots, 4$ is formed for each of the patches A, B, C or D. For example the block A_1 for patch A, with m_1 observations is of the form

$$A_1 = \begin{bmatrix} 1 & v_1 & v_1^2 & v_1^3 & u_1 & u_1 v_1 & u_1 v_1^2 & u_1 v_1^3 & u_1^2 & u_1^2 v_1 & u_1^2 v_1^2 & u_1^2 v_1^3 & u_1^3 & u_1^3 v_1 & u_1^3 v_1^2 & u_1^3 v_1^3 \\ 1 & v_2 & v_2^2 & v_2^3 & u_2 & u_2 v_2 & u_2 v_2^2 & u_2 v_2^3 & u_2^2 & u_2^2 v_2 & u_2^2 v_2^2 & u_2^2 v_2^3 & u_2^3 & u_2^3 v_2 & u_2^3 v_2^2 & u_2^3 v_2^3 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & v_{m_1} & v_{m_1}^2 & v_{m_1}^3 & u_{m_1} & u_{m_1} v_{m_1} & u_{m_1} v_{m_1}^2 & u_{m_1} v_{m_1}^3 & u_{m_1}^2 & u_{m_1}^2 v_{m_1} & u_{m_1}^2 v_{m_1}^2 & u_{m_1}^2 v_{m_1}^3 & u_{m_1}^3 & u_{m_1}^3 v_{m_1} & u_{m_1}^3 v_{m_1}^2 & u_{m_1}^3 v_{m_1}^3 \end{bmatrix} \quad (\text{A.1})$$

and similarly for other patches, A_i , $i = 2, 3, 4$ can be made.

A.2. Submatrices of constraint matrix B in Eq. (17)

The submatrices of constraint matrix B in Eq. (17) are of the form

$$B_1^{(u,v)} = \begin{bmatrix} 1 & v & v^2 & v^3 & u & uv & uv^2 & uv^3 & u^2 & u^2v & u^2v^2 & u^2v^3 & u^3 & u^3v & u^3v^2 & u^3v^3 \\ 0 & 0 & 0 & 0 & 1 & v & v^2 & v^3 & 2u & 2uv & 2uv^2 & 2uv^3 & 3u^2 & 3u^2v & 3u^2v^2 & 3u^2v^3 \\ 0 & 1 & 2v & 3v^2 & 0 & u & 2uv & 3uv^2 & 0 & u^2 & 2u^2v & 3u^2v^2 & 0 & u^3 & 2u^3v & 3u^3v^2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 2v & 2v^2 & 2v^3 & 6u & 6uv & 6uv^2 & 6uv^3 \\ 0 & 0 & 0 & 0 & 0 & 1 & 2v & 3v^2 & 0 & 2u & 4uv & 6uv^2 & 0 & 3u^2 & 6u^2v & 9u^2v^2 \end{bmatrix} \quad (\text{A.2})$$

and

$$B_2^{(u,v)} = \begin{bmatrix} 1 & v & v^2 & v^3 & u & uv & uv^2 & uv^3 & u^2 & u^2v & u^2v^2 & u^2v^3 & u^3 & u^3v & u^3v^2 & u^3v^3 \\ 0 & 0 & 0 & 0 & 1 & v & v^2 & v^3 & 2u & 2uv & 2uv^2 & 2uv^3 & 3u^2 & 3u^2v & 3u^2v^2 & 3u^2v^3 \\ 0 & 1 & 2v & 3v^2 & 0 & u & 2uv & 3uv^2 & 0 & u^2 & 2u^2v & 3u^2v^2 & 0 & u^3 & 2u^3v & 3u^3v^2 \\ 0 & 0 & 2 & 6v & 0 & 0 & 2u & 6uv & 0 & 0 & 2u^2 & 6u^2v & 0 & 0 & 2u^3 & 6u^3v \\ 0 & 0 & 0 & 0 & 0 & 1 & 2v & 3v^2 & 0 & 2u & 4uv & 6uv^2 & 0 & 3u^2 & 6u^2v & 9u^2v^2 \end{bmatrix} \quad (\text{A.3})$$

and

$$B_3^{(u,v)} = \begin{bmatrix} 1 & v & v^2 & v^3 & u & uv & uv^2 & uv^3 & u^2 & u^2v & u^2v^2 & u^2v^3 & u^3 & u^3v & u^3v^2 & u^3v^3 \\ 0 & 1 & 2v & 3v^2 & 0 & u & 2uv & 3uv^2 & 0 & u^2 & 2u^2v & 3u^2v^2 & 0 & u^3 & 2u^3v & 3u^3v^2 \\ 0 & 0 & 2 & 6v & 0 & 0 & 2u & 6uv & 0 & 0 & 2u^2 & 6u^2v & 0 & 0 & 2u^3 & 6u^3v \\ 0 & 0 & 0 & 0 & 0 & 1 & 2v & 3v^2 & 0 & 2u & 4uv & 6uv^2 & 0 & 3u^2 & 6u^2v & 9u^2v^2 \end{bmatrix} \quad (\text{A.4})$$

A.3. Submatrices of constraint matrix B in Eq. (20)

The submatrices of constraint matrix B in Eq. (20) are of the form

$$B_1^{(u,v)} = \begin{bmatrix} 1 & v & v^2 & v^3 & u & uv & uv^2 & uv^3 & u^2 & u^2v & u^2v^2 & u^2v^3 & u^3 & u^3v & u^3v^2 & u^3v^3 \\ 0 & 0 & 0 & 0 & 1 & v & v^2 & v^3 & 2u & 2uv & 2uv^2 & 2uv^3 & 3u^2 & 3u^2v & 3u^2v^2 & 3u^2v^3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 2v & 2v^2 & 2v^3 & 6u & 6uv & 6uv^2 & 6uv^3 \end{bmatrix} \quad (\text{A.5})$$

$$B_2^{(u,v)} = \begin{bmatrix} 1 & v & v^2 & v^3 & u & uv & uv^2 & uv^3 & u^2 & u^2v & u^2v^2 & u^2v^3 & u^3 & u^3v & u^3v^2 & u^3v^3 \\ 0 & 1 & 2v & 3v^2 & 0 & u & 2uv & 3uv^2 & 0 & u^2 & 2u^2v & 3u^2v^2 & 0 & u^3 & 2u^3v & 3u^3v^2 \\ 0 & 0 & 2 & 6v & 0 & 0 & 2u & 6uv & 0 & 0 & 2u^2 & 6u^2v & 0 & 0 & 2u^3 & 6u^3v \end{bmatrix} \quad (\text{A.6})$$

and

$$B_3^{(u,v)} = \begin{bmatrix} 1 & v & v^2 & v^3 & u & uv & uv^2 & uv^3 & u^2 & u^2v & u^2v^2 & u^2v^3 & u^3 & u^3v & u^3v^2 & u^3v^3 \\ 0 & 1 & 2v & 3v^2 & 0 & u & 2uv & 3uv^2 & 0 & u^2 & 2u^2v & 3u^2v^2 & 0 & u^3 & 2u^3v & 3u^3v^2 \\ 0 & 0 & 2 & 6v & 0 & 0 & 2u & 6uv & 0 & 0 & 2u^2 & 6u^2v & 0 & 0 & 2u^3 & 6u^3v \end{bmatrix} \quad (\text{A.7})$$

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